

# Digital Material Passport

ID **DMP-METAL-012** - Version **1.0.0**  
Issue Date **2025-05-14** - Certificate Type **EN 10204 3.1**

| Manufacturer                                                                                         | Customer                                                                                                             | Business Transaction |                                                   |
|------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|----------------------|---------------------------------------------------|
| ACME Metal Works GmbH<br>Industrial Park 123<br>52066 Aachen<br>DE<br>quality@acme-metal.example.com | Engineering Solutions Ltd.<br>Tech Park Way 45<br>Cardiff<br>CF14 5DU<br>GB<br>procurement@engisolutions.example.com | Order                | PO-78901 · <b>5000 kg</b><br>Pos. 10 · 2025-04-20 |
|                                                                                                      |                                                                                                                      | Delivery             | DN-56789 · <b>5000 kg</b><br>Pos. 1 · 2025-05-12  |

## Product

|                   |                                    |              |                                                                                                                                                                     |
|-------------------|------------------------------------|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Product Name      | Structural Steel S355J2            | Name (EN)    | 1.0577                                                                                                                                                              |
| Batch ID          | H-10987-02                         | Shape        | RoundBar - D 50 mm - Length 6000 mm                                                                                                                                 |
| Strand ID         | 3                                  | Packaging    | Hexagonal bundles of 5 pieces with wire binding in good condition.                                                                                                  |
| Billet Number     | B-4471-01                          | Product Norm | EN 10025-2 (2019)                                                                                                                                                   |
| Billet Number     | B-4471-02                          |              | 720839 - Flat-rolled products of iron or non-alloy steel, of a width of 600 mm or more, hot-rolled, not clad, plated or coated                                      |
| Bundle Number     | BDL-118                            | HS Code      | 72083900 - Flat-rolled products of iron or non-alloy steel, of a width of 600 mm or more, hot-rolled, not clad, plated or coated, of a thickness of 4.75 mm or more |
| Surface Condition | Hot-rolled                         | CN8 (EU)     |                                                                                                                                                                     |
| Weight            | 5000 kg                            |              |                                                                                                                                                                     |
| Production Date   | 2025-05-09                         |              |                                                                                                                                                                     |
| Country of Origin | DE                                 |              |                                                                                                                                                                     |
| Specification     | EN 10025-2 - Rev 2019 - 2019-11-01 |              |                                                                                                                                                                     |

## Heat Treatment

Process: **Normalizing** - Lot: HT-2024-11-15-B47 - Furnace: FURNACE-03 - Date: 2024-11-15

| Stage         | Temperature | Duration | Cooling | Atmosphere |
|---------------|-------------|----------|---------|------------|
| Austenitizing | 920 C       |          |         |            |

## Chemical Analysis

Heat Number **H-10987** - Melting Process **EAF+LF** - Casting Method **ContinuousCasting** - Casting Date **2025-05-08** - Sample Location **Ladle**

| Symbol | C    | Mn   | Si   | P     | S     | N     | CEV  |
|--------|------|------|------|-------|-------|-------|------|
| Unit   | %    | %    | %    | %     | %     | %     | %    |
| Min    |      |      |      |       |       |       |      |
| Max    | 0.20 | 1.60 | 0.50 | 0.025 | 0.020 | 0.009 | 0.45 |
| Actual | 0.18 | 1.45 | 0.25 | 0.018 | 0.012 | 0.006 | 0.42 |

CEV = C+Mn/6+(Cr+Mo+V)/5+(Ni+Cu)/15: 0.42%

## Mechanical Properties

| Tensile Strength <sup>1</sup>          | Symbol | 1    | 2   | 3    | Average | Min  | Max  | Spec Min | Spec Max | Method        |
|----------------------------------------|--------|------|-----|------|---------|------|------|----------|----------|---------------|
| Value [MPa]                            | Rm     | 508  | 510 | 512  | 510     | 508  | 512  | 470      | 630      | EN ISO 6892-1 |
|                                        |        |      |     |      |         |      |      |          |          |               |
| Yield Strength <sup>1</sup>            | Symbol | 1    | 2   | 3    | Average | Min  | Max  | Spec Min | Spec Max | Method        |
| Value [MPa]                            | ReH    | 378  | 380 | 382  | 380     | 378  | 382  | 355      | 630      | EN ISO 6892-1 |
|                                        |        |      |     |      |         |      |      |          |          |               |
| Elongation after fracture <sup>1</sup> | Symbol | 1    | 2   | 3    | Average | Min  | Max  | Std Dev  | Spec Min | Method        |
| Value [%]                              | A      | 21.5 | 22  | 22.5 | 22      | 21.5 | 22.5 | 0.5      | 20       | EN ISO 6892-1 |

<sup>1</sup> 3 specimens tested

## Validation

We hereby certify that the material described above has been manufactured and tested in accordance with the requirements of EN 10204:2004 type 3.1 and the specified standards. The results comply with the requirements.

Validated by **Johann Weber**, Quality Inspector, Quality Assurance - 2025-05-14